CISCO SECURE · HMF DESIGN CLINIC

Team Workbook

Cedarline Health Group, a Hybrid Mesh Firewall design mission

Today your team is the Cisco Solution Engineering team walking into Cedarline. You will sit with four real conversations, hear what is keeping them up at night, and answer as Solution Engineers do, with the right solution, why it fits, and the outcome it delivers, all brought together into one Hybrid Mesh Firewall design.

The scenario. Cedarline Health Group is a regional healthcare payer-provider with roughly 8,000 employees across four hospitals and thirty-one clinics, plus a claims-processing business that handles the money. Growth by acquisition (including a second data centre absorbed as-is) and a mid-flight, disorderly move to public cloud have left a fragmented security posture. After a security incident in March, the board asked what would have stopped it, and leadership still cannot answer. Budget and executive attention are approved but explicitly temporary, and the security team is small.

TEAM	MEMBERS (UP TO 8)	DATE

START HERE

What you will walk away able to do

1. Map Cedarline's stated concerns to the right Cisco Secure solutions.
2. Explain, in the customer's language, how each solution addresses the concern.
3. Connect the four case studies into one coherent Hybrid Mesh Firewall design.
4. Present the business outcome, not just the feature.

How your answers are assessed

Each case study is scored across its four pain points, giving a case study score and an overall score for the clinic. Alongside the score, your answers are assessed on three dimensions.

Problem Identifier

Correctly identified the customer's pain point and the requirements behind it.

Solution Designer

Recommended the right Cisco solution, architecture, and approach.

Value Communicator

Clearly explained how the solution works and the customer outcomes it enables.

The two-part answer. For every customer concern you will write a **Product / platform** and a **Solution Engineer response** that covers how it addresses the concern and the outcome it delivers. Identify the problem, design the right solution, and communicate the value.

The Cisco Secure portfolio, at a glance

A quick decoder of the products in play. Which product fits which concern is for you to decide.

Cisco AI Defense	Protects AI apps and models with validation, runtime guardrails, supply-chain scanning, and AI visibility across clouds.
Cisco Multicloud Defense	Gives one security policy plane across AWS, Azure and GCP, with segmentation, egress control, and unified visibility.
Cisco Secure Firewall (FTD) + FMC	The next-generation firewall and its manager, the Firewall Management Center.
Cisco Security Cloud Control	Cloud-delivered management across the whole firewall estate.
Cisco Secure Workload	Maps how applications actually talk, then drives safe micro-segmentation.
Cisco ISE	The Identity Services Engine that knows who and what is on the network and can quarantine fast.
Isovalent Cilium	eBPF-based Kubernetes networking and policy.
Isovalent Tetragon	eBPF-based runtime security and threat detection for workloads.

Decode the jargon

The terms you will hear Cedarline use, in plain English.

EU AI Act, Article 15	EU law classifying AI by risk. A 'Provider' must prove robustness and cybersecurity.
MCP (Model Context Protocol)	How AI agents connect to tools and data, creating a new supply-chain surface to secure.
East-west traffic	Traffic between workloads inside the data centre, as opposed to north-south, which flows in and out.
Lateral movement	An attacker hopping from one compromised system to others on the inside.
Micro-segmentation	Fine-grained isolation of individual workloads so a breach cannot spread.
eBPF	Linux kernel technology used to observe and enforce networking and security with low overhead.
EVE (Encrypted Visibility Engine)	Infers threats from encrypted traffic patterns without decrypting it.
Snort ML	Machine-learning detection inside the Snort engine for unknown and zero-day exploits.

pxGrid	Cisco's context-sharing fabric so tools can act on identity, for example an automatic quarantine via ISE.
Hybrid Mesh Firewall (HMF)	Consistent security enforcement across firewalls, cloud gateways, workloads and Kubernetes, managed centrally.

1 Case Study 1

The conversation. Cedarline is standing up AI across regulated workflows and public cloud faster than security can see it. Help them prove compliance, vet what they adopt, and get visibility and guardrails over AI traffic.

PAIN POINT 1 Case Study 1

> *Under the EU AI Act, we are classified as a 'Provider' for our internal AI systems. We need to demonstrate robustness and cybersecurity compliance to satisfy Article 15. How does Cisco help us meet these specific requirements?*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 2 Case Study 1

> *We are integrating various open-source models and third-party agents into our internal workflows. How can we ensure these assets aren't introducing vulnerabilities into our environment before they even go live?*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 3 Case Study 1

> *We know AI is showing up across our cloud environments faster than security can track it. We're worried we have models, agents, datasets, and MCP-connected tools running without visibility or consistent controls.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 4 Case Study 1

> *We're running AI workloads in cloud accounts, but we don't have a reliable way to see which models and AI assets are actually in use. We're also concerned about workloads reaching out to third-party AI models without security knowing about it. And even when we do know the traffic path, we need a practical way to inspect and enforce guardrails on prompt and response traffic in the cloud without changing every application.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

2 Case Study 2

The conversation. Workloads sprawl across AWS, Azure and on-prem, each with its own policy model. Contain lateral movement, prove consistent controls to auditors, and give Cedarline one place to define and enforce policy.

PAIN POINT 1 Case Study 2

> *If a breach occurs in one of our public cloud instances, we are terrified that the attacker will move laterally into our core on-premises systems. How can we contain these threats in a multicloud environment?*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 2 Case Study 2

> *Our auditors require proof that our security controls are applied consistently across all our cloud workloads. Gathering this data from different cloud providers is a manual, time-consuming nightmare.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 3 Case Study 2

> *We have workloads spread across AWS, Azure, and on-premises data centers. Managing security policies individually in each cloud environment is creating massive operational overhead and leading to inconsistent security posture.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 4 Case Study 2

> *We have workloads on-prem, in cloud, and in mixed environments, and policy consistency is becoming difficult. Every environment feels different, and we're worried about drift.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

3 Case Study 3

The conversation. A mixed-vendor perimeter, an inherited rulebase and mostly encrypted traffic leave blind spots and a slow response. Restore visibility, speed the response, and modernise detection with the team they have.

PAIN POINT 1 Case Study 3

> *When our security tools detect a compromised device, our manual response time is too slow. By the time we identify the user and manually isolate the port, the attacker has already moved through the network.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 2 Case Study 3

> *Our firewall environment has become too complex to manage manually.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 3 Case Study 3

> *Most of our traffic is encrypted, so we've lost visibility into a huge part of our environment.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 4 Case Study 3

> *We're worried traditional signature-based detection won't catch new or unknown attacks quickly enough.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

4 Case Study 4

The conversation. A flat data centre and fragmented Kubernetes networking invite east-west movement. Build segmentation Cedarline can adopt safely, with runtime visibility and application-dependency awareness so nothing breaks.

PAIN POINT 1 Case Study 4

> *Network policy helps, but we also need to know what's happening at runtime inside the workload. We need better detection of suspicious behavior and policy violations.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 2 Case Study 4

> *Our Kubernetes networking feels fragmented across clusters and environments. We're dealing with too many moving parts, inconsistent policies, and too much operational complexity.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 3 Case Study 4

> *Right now we're stitching together separate tools for networking, security, and observability in Kubernetes, and it's creating too much overhead.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

PAIN POINT 4 Case Study 4

> *We know we need microsegmentation, but we don't have enough visibility into east-west flows or application dependencies to create policy safely. We don't want to break the application.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

Your Hybrid Mesh Firewall design

Four conversations, one story. Combine your case studies into a single HMF design and paste your team's topology below, then be ready to walk Cedarline through it like the trusted advisors you are.

Paste your team's topology diagram here

Show how the four case studies connect across Cedarline's on-prem, data-centre and multicloud estate.